

E-commerce Microservices Jenkins Integration Project

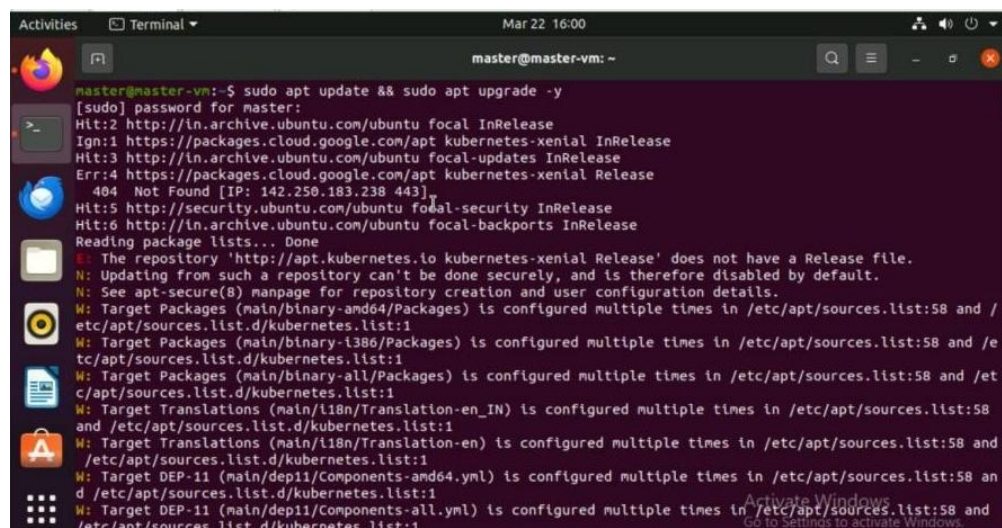
Prerequisites

System Requirements

- **Ubuntu (or any Linux distribution):** Used as the base operating system for running applications and services in a stable, secure, and scalable environment.
- **Docker & Docker Compose:** Used for containerizing applications and managing multicontainer environments efficiently.
- **Java 21:** Provides the runtime and development environment for running modern Java applications with long-term support.
- **Gradle 8.5+:** Used as a build automation tool for managing dependencies, compiling code, and packaging applications.
- **PostgreSQL (optional if using Docker):** Serves as a reliable and powerful relational database for storing application data.
- **Kafka:** Used as a distributed event streaming platform for building real-time data pipelines and event-driven architectures.

Step 1: Install Required Software

Update System Packages



```
master@master-vm: ~  
master@master-vm:~$ sudo apt update && sudo apt upgrade -y  
[sudo] password for master:  
Hit:2 http://ln.archive.ubuntu.com/ubuntu focal InRelease  
Ign:1 https://packages.cloud.google.com/apt kubernetes-xenial InRelease  
Hit:3 http://ln.archive.ubuntu.com/ubuntu focal-updates InRelease  
Err:4 https://packages.cloud.google.com/apt kubernetes-xenial Release  
404 Not Found [IP: 142.250.183.238 443]  
Hit:5 http://security.ubuntu.com/ubuntu focal-security InRelease  
Hit:6 http://ln.archive.ubuntu.com/ubuntu focal-backports InRelease  
Reading package lists... Done  
E: The repository 'http://apt.kubernetes.io kubernetes-xenial Release' does not have a Release file.  
N: Updating from such a repository can't be done securely, and is therefore disabled by default.  
N: See apt-secure(8) manpage for repository creation and user configuration details.  
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target Packages (main/binary-i386/Packages) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target Translations (main/i18n/Translation-en_IN) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1  
W: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list:58 and /etc/apt/sources.list.d/kubernetes.list:1
```

```
Activities Terminal Mar 22 16:00
master@master-vm: ~
s.list:1
master@master-vm:~$ sudo apt install openjdk-21-jdk -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev
openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless
xi1proto-core-dev xi1proto-dev xorg-sgml-doctools xtrans-dev
Suggested packages:
default-jre libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc
openjdk-21-demo openjdk-21-source visualvm fonts-ipafont-gothic
fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei
The following NEW packages will be installed:
ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-21-jdk
openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless
xi1proto-core-dev xi1proto-dev xorg-sgml-doctools xtrans-dev
0 upgraded, 21 newly installed, 0 to remove and 2 not upgraded.
Need to get 130 MB of archives.
After this operation, 316 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu focal/main amd64 java-common all 0.72 [6,816 B]
Get:2 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 openjdk-21-jre-headless amd64 21.0.6+7-1~20.04.1 [41.6 MB]
Get:3 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 ca-certificates-java all 20190405ubuntu1.1 [12.4 KB]
```

Step 2: Setting up Gradle Environment.

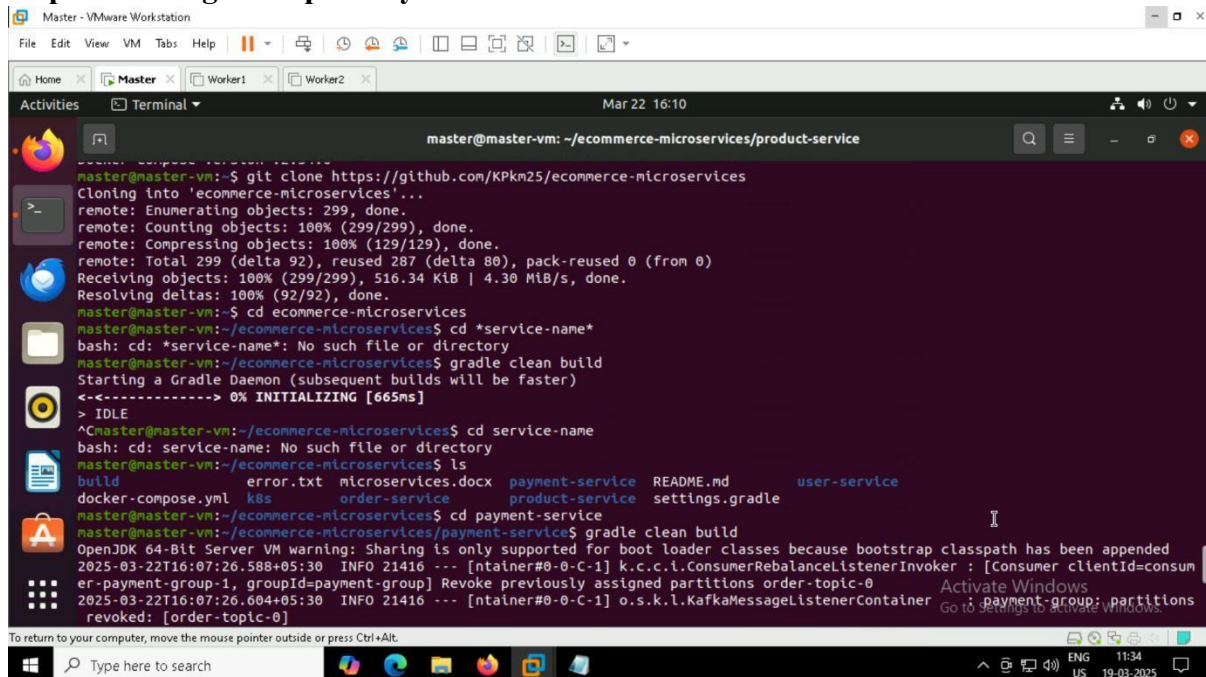
```
Activities Terminal Mar 22 16:01
master@master-vm: ~
--2025-03-22 15:57:15-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/690192900/uc09afce-7970-4c71-9
704-6dcd18882ea?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250322%2Fus-east-1%2Fs3%2Faws4_request
&X-Amz-Date=20250322T102606Z&X-Amz-Expires=300&X-Amz-Signature=d3bc9d4fea799f7c0354ace594373dead963b2ad6bbb5ab965a84d1298b418b2&X-Am
z-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dgradle-8.5-bin.zip&response-content-type=application%2F
octet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.111.133, 185.199.109.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 132519731 (126M) [application/octet-stream]
Saving to: '/tmp/gradle-8.5-bin.zip'

gradle-8.5-bin.zip 100%[=====] 126.38M 37.0MB/s in 3.8s

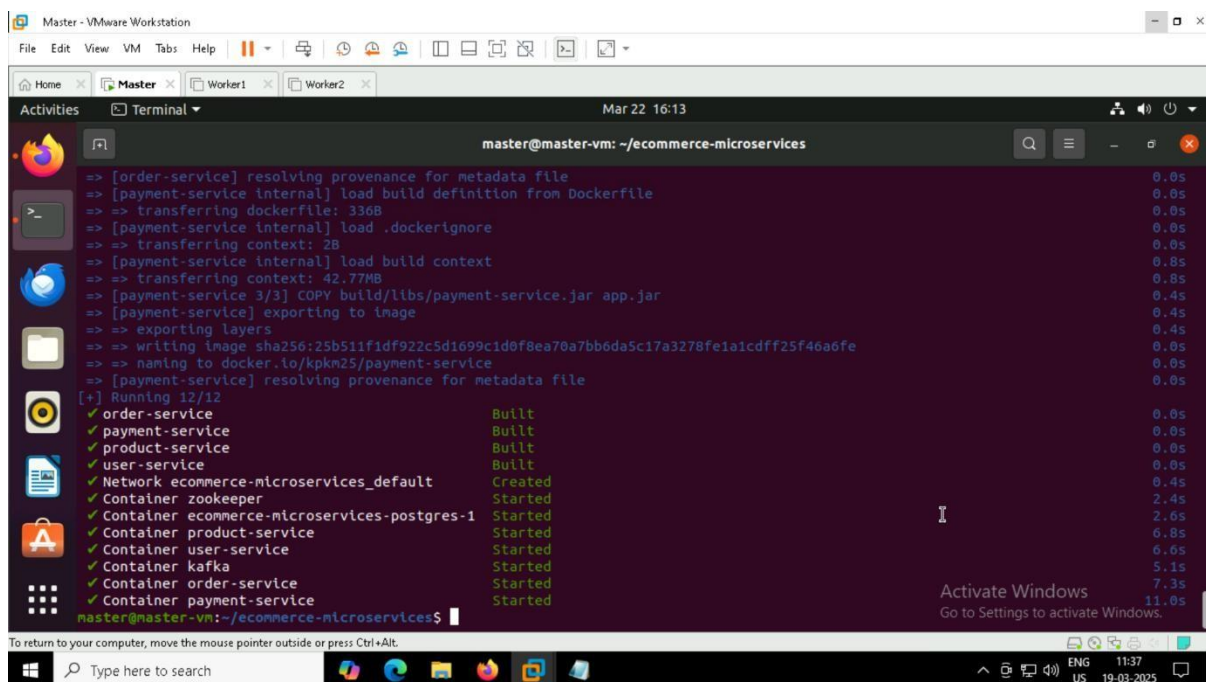
2025-03-22 15:57:18 (32.9 MB/s) - '/tmp/gradle-8.5-bin.zip' saved [132519731/132519731]

master@master-vm:~$ sudo unzip -d /opt/gradle /tmp/gradle-8.5-bin.zip
Archive: /tmp/gradle-8.5-bin.zip
creating: /opt/gradle/gradle-8.5/
inflating: /opt/gradle/gradle-8.5/LICENSE
inflating: /opt/gradle/gradle-8.5/NOTICE
inflating: /opt/gradle/gradle-8.5/README
creating: /opt/gradle/gradle-8.5/init.d/
inflating: /opt/gradle/gradle-8.5/init.d/readme.txt
creating: /opt/gradle/gradle-8.5/bin/
inflating: /opt/gradle/gradle-8.5/bin/gradle
inflating: /opt/gradle/gradle-8.5/bin/gradle.bat
creating: /opt/gradle/gradle-8.5/lib/
```

Step 3: Cloning the repository.



```
master@master-vm: ~/ecommerce-microservices/product-service
master@master-vm:~$ git clone https://github.com/KPkm25/ecommerce-microservices
Cloning into 'ecommerce-microservices'...
remote: Enumerating objects: 299, done.
remote: Counting objects: 100% (299/299), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 299 (delta 92), reused 287 (delta 80), pack-reused 0 (from 0)
Receiving objects: 100% (299/299), 516.34 KiB | 4.30 MiB/s, done.
Resolving deltas: 100% (92/92), done.
master@master-vm:~$ cd ecommerce-microservices
master@master-vm:~/ecommerce-microservices$ cd *service-name*
bash: cd: *service-name*: No such file or directory
master@master-vm:~/ecommerce-microservices$ gradle clean build
Starting a Gradle Daemon (subsequent builds will be faster)
<<-----> 0% INITIALIZING [665ms]
> IDLE
^Cmaster@master-vm:~/ecommerce-microservices$ cd service-name
bash: cd: service-name: No such file or directory
master@master-vm:~/ecommerce-microservices$ ls
build          error.txt      microservices.docx  payment-service  README.md       user-service
docker-compose.yml  k8s           order-service       product-service  settings.gradle
master@master-vm:~/ecommerce-microservices$ cd payment-service
master@master-vm:~/ecommerce-microservices/payment-service$ gradle clean build
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
2025-03-22T16:07:26.588+05:30 INFO 21416 --- [ntainer#0-0-C-1] k.c.c.i.ConsumerRebalanceListenerInvoker : [Consumer clientId=consum
er-payment-group-1, groupId=payment-group] Revoke previously assigned partitions order-topic-0
2025-03-22T16:07:26.604+05:30 INFO 21416 --- [ntainer#0-0-C-1] o.s.k.l.KafkaMessageListenerContainer : payment-group: partitions
revoked: [order-topic-0]
```



```
master@master-vm:~/ecommerce-microservices
=> [order-service] resolving provenance for metadata file 0.0s
=> [payment-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 336B 0.0s
=> [payment-service internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [payment-service internal] load build context 0.8s
=> => transferring context: 42.77MB 0.8s
=> [payment-service 3/3] COPY build/libs/payment-service.jar app.jar 0.4s
=> [payment-service] exporting to image 0.4s
=> => exporting layers 0.4s
=> => writing image sha256:25b511f1df922c5d1699c1d0f8ea70a7bb6da5c17a3278fe1a1cdf25f46a6fe 0.0s
=> => naming to docker.io/kpkm25/payment-service 0.0s
=> [payment-service] resolving provenance for metadata file 0.0s
[+] Running 12/12
✔ order-service Built 0.0s
✔ payment-service Built 0.0s
✔ product-service Built 0.0s
✔ user-service Built 0.0s
✔ Network ecommerce-microservices_default Created 0.4s
✔ Container zookeeper Started 2.4s
✔ Container ecommerce-microservices-postgres-1 Started 2.6s
✔ Container product-service Started 6.8s
✔ Container user-service Started 6.6s
✔ Container kafka Started 5.1s
✔ Container order-service Started 7.3s
✔ Container payment-service Started 11.0s
master@master-vm:~/ecommerce-microservices$
```



```
master@master-vm: ~/ecommerce-microservices

BUILD SUCCESSFUL in 19s
9 actionable tasks: 8 executed, 1 up-to-date
master@master-vm:~/ecommerce-microservices/product-service$ cd ..
master@master-vm:~/ecommerce-microservices$ gradle clean build -x test
Path for java installation '/usr/lib/jvm/openjdk-21' (Common Linux Locations) does not contain a java executable

BUILD SUCCESSFUL in 16s
23 actionable tasks: 23 executed
master@master-vm:~/ecommerce-microservices$ docker-compose up -d --build
WARN[0000] /home/master/ecommerce-microservices/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please
remove it to avoid potential confusion
[+] Running 34/34
  ✓ kafka Pulled                                89.2s
  ✓ postgres Pulled                             52.5s
  ✓ zookeeper Pulled                             90.1s
Compose can now delegate builds to bake for better performance.
To do so, set COMPOSE_BAKE=true.
[+] Building 22.3s (11/16)
=> => transferring dockerfile: 333B                                docker:default 0.0s
=> [order-service auth] library/openjdk:pull token for registry-1.docker.io 0.0s
=> [user-service internal] load .dockerignore                        0.0s
=> => transferring context: 2B                                       0.0s
=> [order-service internal] load .dockerignore                      0.1s
=> => transferring context: 2B                                       0.0s
=> [product-service internal] load .dockerignore                    0.1s
=> => transferring context: 2B                                       0.0s
```

```
Master x Worker1 x Worker2 x
Activities Terminal Mar 22 16:01
master@master-vm: ~

inflating: /opt/gradle/gradle-8.5/lib/plugins/opencensus-api-0.31.1.jar
inflating: /opt/gradle/gradle-8.5/lib/plugins/grpc-context-1.27.2.jar
creating: /opt/gradle/gradle-8.5/lib/agents/
inflating: /opt/gradle/gradle-8.5/lib/agents/gradle-instrumentation-agent-8.5.jar
master@master-vm:~$ echo 'export PATH=/opt/gradle/gradle-8.5/bin:$PATH' >> ~/.bashrc
master@master-vm:~$ source ~/.bashrc
master@master-vm:~$ gradle -v

Welcome to Gradle 8.5!

Here are the highlights of this release:
- Support for running on Java 21
- Faster first use with Kotlin DSL
- Improved error and warning messages

For more details see https://docs.gradle.org/8.5/release-notes.html

-----
Gradle 8.5
-----
Build time: 2023-11-29 14:08:57 UTC
Revision: 28aca86a7180baa17117e0e5ba01d8ea9feca598

Kotlin: 1.9.20
Groovy: 3.0.17
```

Step 4: Running the Application

Start the Services with Docker Compose

Open docker-compose and change the image tag to your Docker Hub credentials. Run

```

Master - VMware Workstation
File Edit View VM Tabs Help
Master Worker1 Worker2
Activities Terminal Mar 22 16:16
master@master-vm: ~/ecommerce-microservices

remove it to avoid potential confusion
[+] Running 8/8
✓ Network ecommerce-microservices_default Created 0.2s
✓ Container zookeeper Started 2.6s
✓ Container ecommerce-microservices-postgres-1 Started 2.7s
✓ Container product-service Started 7.9s
✓ Container user-service Started 7.9s
✓ Container kafka Started 6.3s
✓ Container order-service Started 8.9s
✓ Container payment-service Started 12.2s

master@master-vm:~/ecommerce-microservices$ docker exec -it ecommerce-microservices-postgres-1 psql -U postgres
psql (17.4 (Debian 17.4-1.pgdg120+2))
Type "help" for help.

postgres=# \l

```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
postgres	postgres	UTF8	libc	en_US.utf8	en_US.utf8			
template0	postgres	UTF8	libc	en_US.utf8	en_US.utf8			=c/postgres postgres=Ctc/postgres +
template1	postgres	UTF8	libc	en_US.utf8	en_US.utf8			=c/postgres postgres=Ctc/postgres +

```

(3 rows)

postgres=# CREATE DATABASE ecommerce;
CREATE DATABASE

```

```

Master - VMware Workstation
File Edit View VM Tabs Help
Master Worker1 Worker2
Activities Terminal Mar 22 16:13
master@master-vm: ~/ecommerce-microservices

=> [order-service] resolving provenance for metadata file 0.0s
=> [payment-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 336B 0.0s
=> [payment-service internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [payment-service internal] load build context 0.8s
=> => transferring context: 42.77MB 0.8s
=> [payment-service 3/3] COPY build/libs/payment-service.jar app.jar 0.4s
=> [payment-service] exporting to image 0.4s
=> => writing image sha256:25b511f1df922c5d1699c1d0f8ea70a7bb6da5c17a3278fe1a1cdf25f46a6fe 0.0s
=> => naming to docker.io/kpkm25/payment-service 0.0s
=> [payment-service] resolving provenance for metadata file 0.0s
[+] Running 12/12
✓ order-service Built 0.0s
✓ payment-service Built 0.0s
✓ product-service Built 0.0s
✓ user-service Built 0.0s
✓ Network ecommerce-microservices_default Created 0.4s
✓ Container ecommerce-microservices-postgres-1 Started 2.6s
✓ Container product-service Started 6.8s
✓ Container user-service Started 6.6s
✓ Container kafka Started 5.1s
✓ Container order-service Started 7.3s
✓ Container payment-service Started 11.0s

master@master-vm:~/ecommerce-microservices$

```

Step 5: Kubernetes Setup

Push the built images to Docker Hub using the following commands:

[illegible]

Master - VMware Workstation

File Edit View VM Tabs Help

Home Master Worker1 Worker2

Activities Terminal Mar 22 16:16

master@master-vm: ~/ecommerce-microservices

```

-----+-----+-----+-----+-----+
public | products | table | postgres
public | users    | table | postgres
(2 rows)

postgres=# ^C
postgres=# exit
master@master-vm:~/ecommerce-microservices$ docker-compose build
WARN[0000] /home/master/ecommerce-microservices/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please
remove it to avoid potential confusion
Compose can now delegate builds to bake for better performance.
To do so, set COMPOSE_BAKE=true.
[+] Building 2.5s (27/27) FINISHED                                docker:default
=> [product-service internal] load build definition from Dockerfile                                0.0s
=> == transferring dockerfile: 336B                                                                0.0s
=> [payment-service internal] load metadata for docker.io/library/openjdk:21-jdk-slim              1.7s
=> [user-service internal] load build definition from Dockerfile                                0.0s
=> == transferring dockerfile: 333B                                                                0.0s
=> [order-service internal] load build definition from Dockerfile                                0.0s
=> == transferring dockerfile: 334B                                                                0.0s
=> [order-service internal] load .dockerignore                                                    0.0s
=> == transferring context: 2B                                                                      0.0s
=> [user-service internal] load .dockerignore                                                    0.0s
=> == transferring context: 2B                                                                      0.0s
=> [product-service internal] load .dockerignore                                                    0.0s
=> == transferring context: 2B                                                                      0.0s
=> [order-service internal] load build context                                                    0.0s

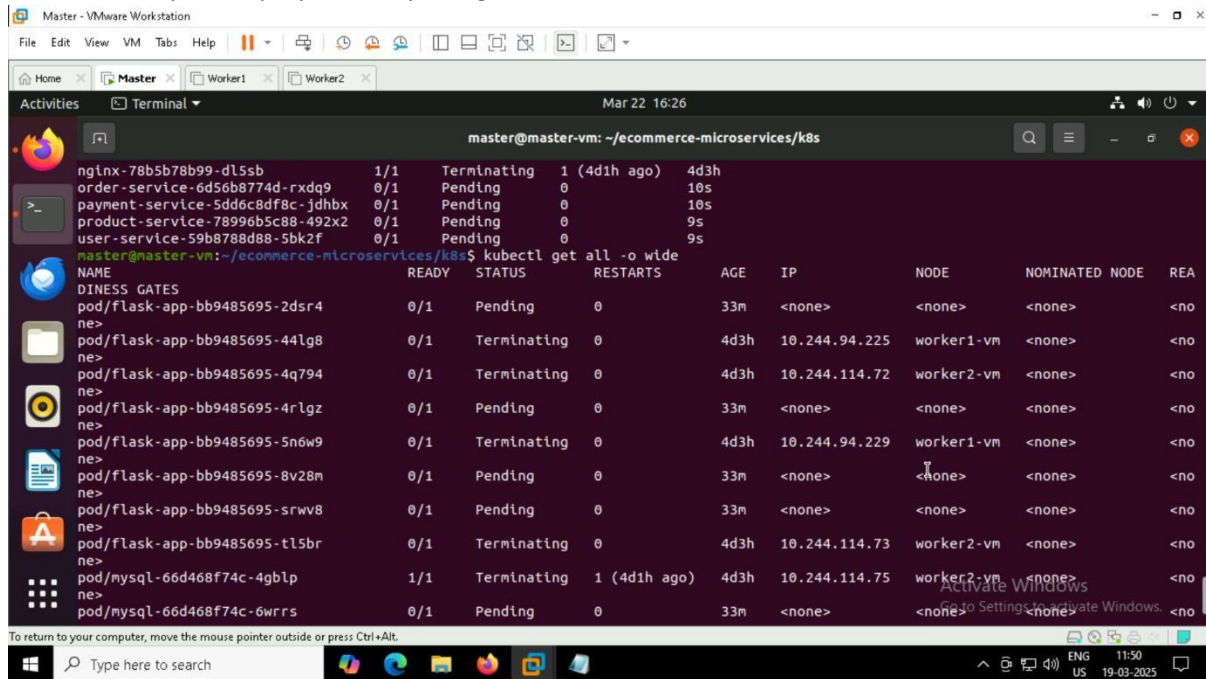
```

Activate Windows
Go to Settings to activate Windows

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

Step 6: Deployment.

You can view your deployments by using:



The screenshot shows a terminal window titled "master@master-vm: ~/ecommerce-microservices/k8s". The terminal displays the output of the command `kubectl get all -o wide`. The output lists various Kubernetes resources including services, pods, and their status. The terminal also shows a list of pods with their names, namespaces, and statuses. The terminal window is part of a VMware Workstation interface, with tabs for "Master", "Worker1", and "Worker2". The terminal output is as follows:

```
master@master-vm: ~/ecommerce-microservices/k8s$ kubectl get all -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	REA
pod/flask-app-bb9485695-2dsr4	0/1	Pending	0	33m	<none>	<none>	<none>	<no	<no
pod/flask-app-bb9485695-44lg8	0/1	Terminating	0	4d3h	10.244.94.225	worker1-vm	<none>	<no	<no
pod/flask-app-bb9485695-4q794	0/1	Terminating	0	4d3h	10.244.114.72	worker2-vm	<none>	<no	<no
pod/flask-app-bb9485695-4rlgz	0/1	Pending	0	33m	<none>	<none>	<none>	<no	<no
pod/flask-app-bb9485695-5n6w9	0/1	Terminating	0	4d3h	10.244.94.229	worker1-vm	<none>	<no	<no
pod/flask-app-bb9485695-8v28m	0/1	Pending	0	33m	<none>	<none>	<none>	<no	<no
pod/flask-app-bb9485695-srwv8	0/1	Pending	0	33m	<none>	<none>	<none>	<no	<no
pod/flask-app-bb9485695-tl5br	0/1	Terminating	0	4d3h	10.244.114.73	worker2-vm	<none>	<no	<no
pod/mysql-66d468f74c-4gblp	1/1	Terminating	1 (4d1h ago)	4d3h	10.244.114.75	worker2-vm	<none>	<no	<no
pod/mysql-66d468f74c-6wrrs	0/1	Pending	0	33m	<none>	<none>	<none>	<no	<no